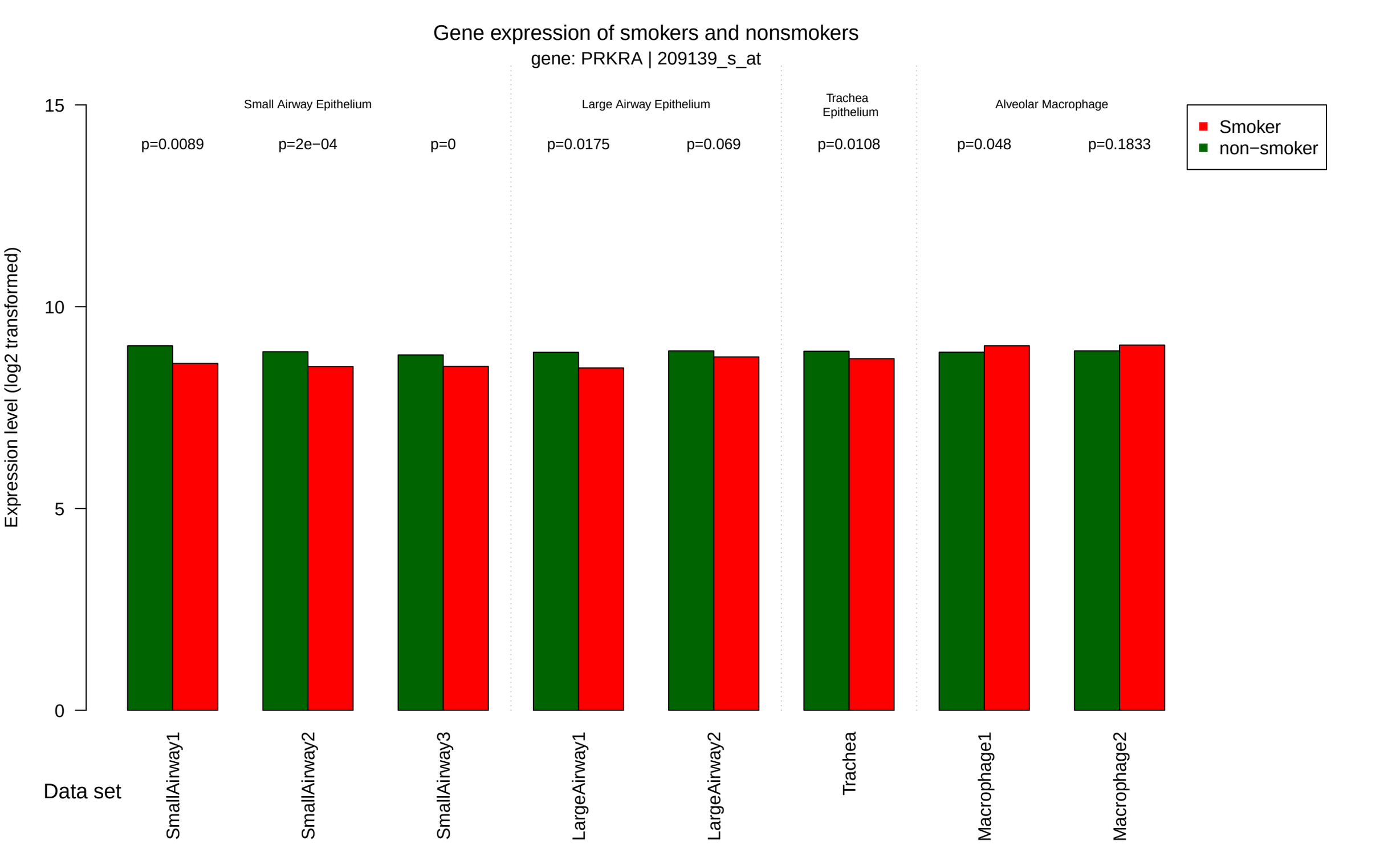
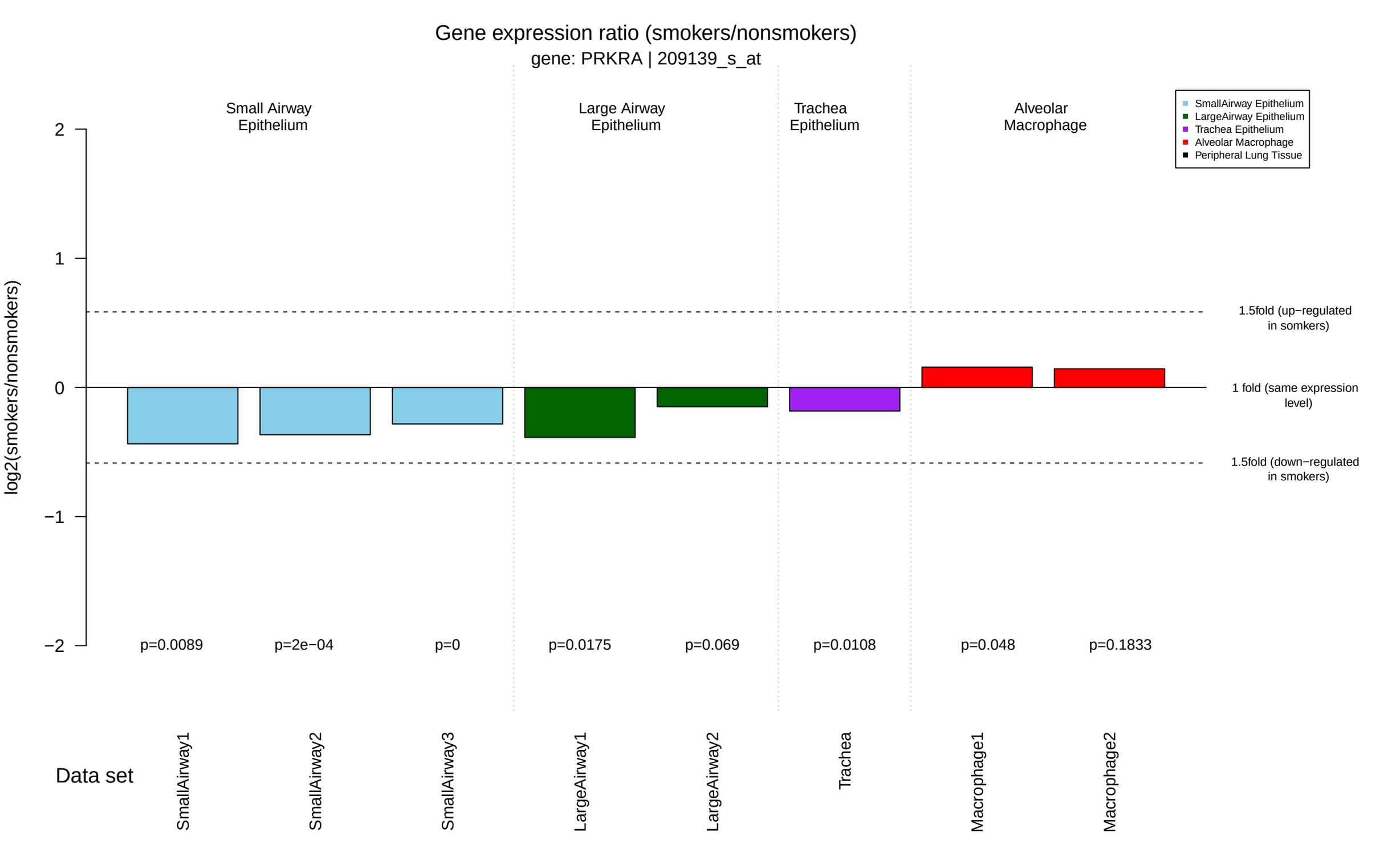
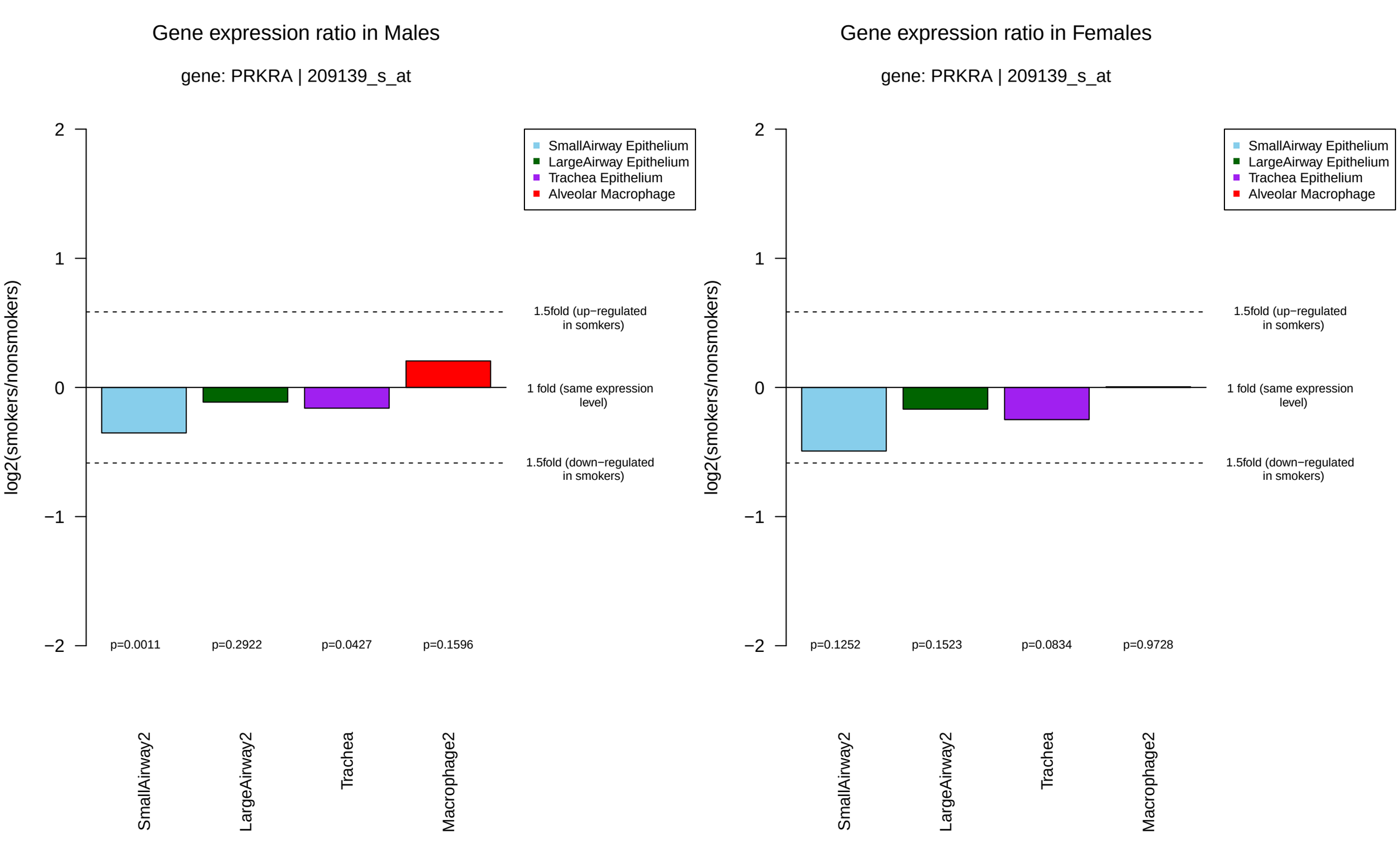
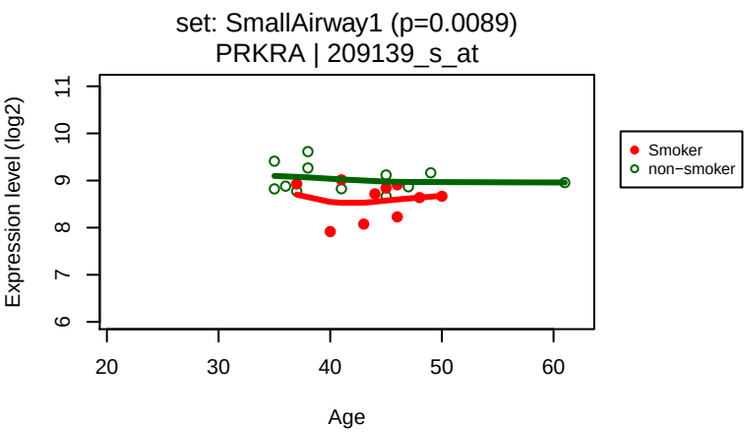
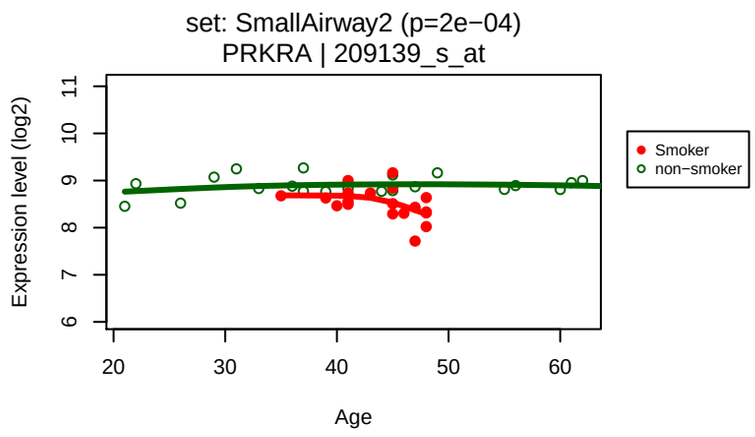
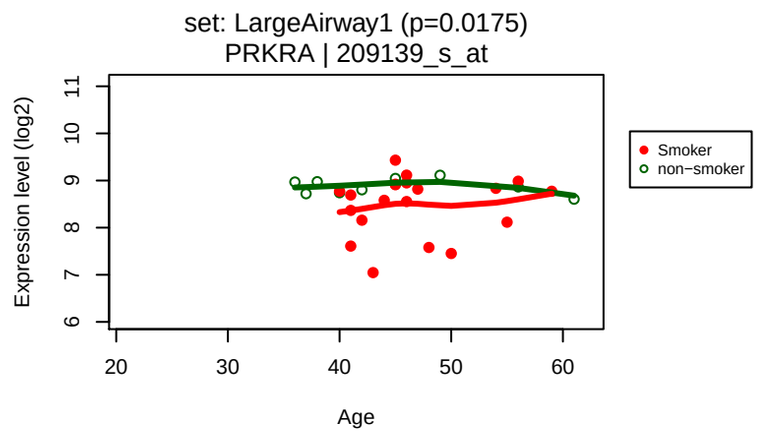

Gene expression of smokers and nonsmokers
gene: PRKRA | 209139_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0089
p=2e−04
p=0
p=0.0175
p=0.069
p=0.0108
p=0.048
p=0.1833
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

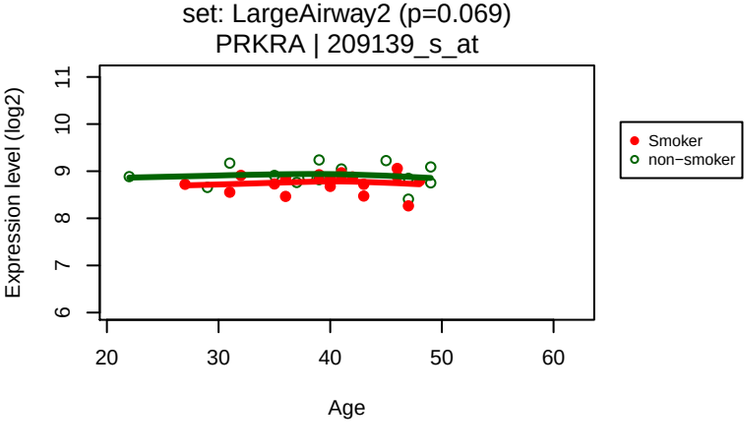

Gene expression ratio (smokers/nonsmokers)
gene: PRKRA | 209139_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0089
p=2e−04
p=0
p=0.0175
p=0.069
p=0.0108
p=0.048
p=0.1833
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

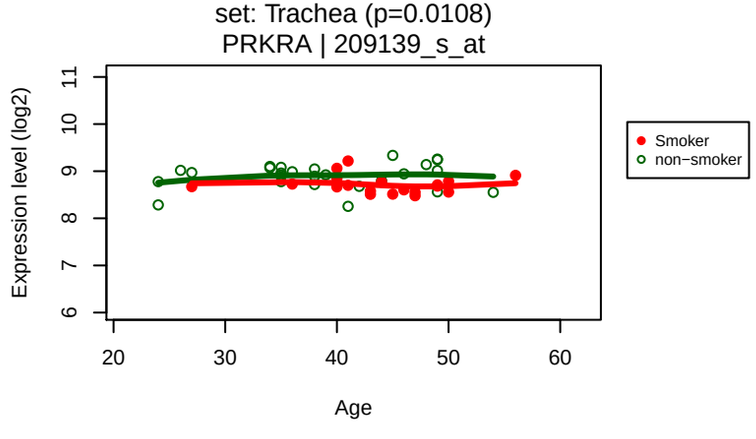

Gene expression ratio in Males
gene: PRKRA | 209139_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0011
p=0.2922
p=0.0427
p=0.1596
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRKRA | 209139_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1252
p=0.1523
p=0.0834
p=0.9728
SmallAirway2
LargeAirway2
Trachea
Macrophage2

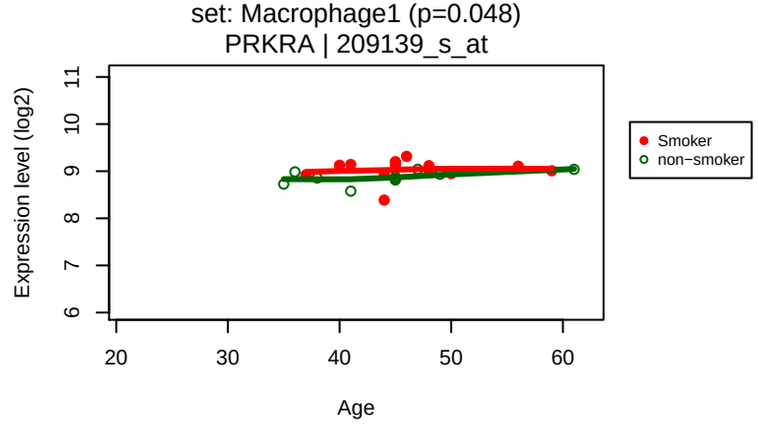

set: SmallAirway1 (p=0.0089)
PRKRA | 209139_s_at
Expression level (log2)
Age
Smoker
non−smoker

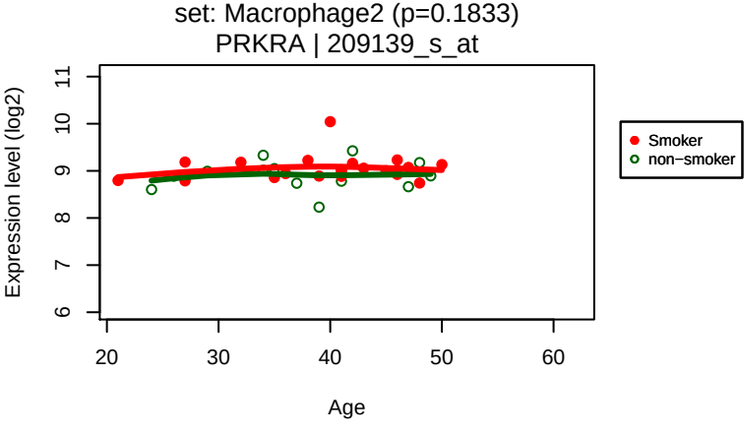

set: SmallAirway2 (p=2e−04)
PRKRA | 209139_s_at
Expression level (log2)
Age
Smoker
non−smoker

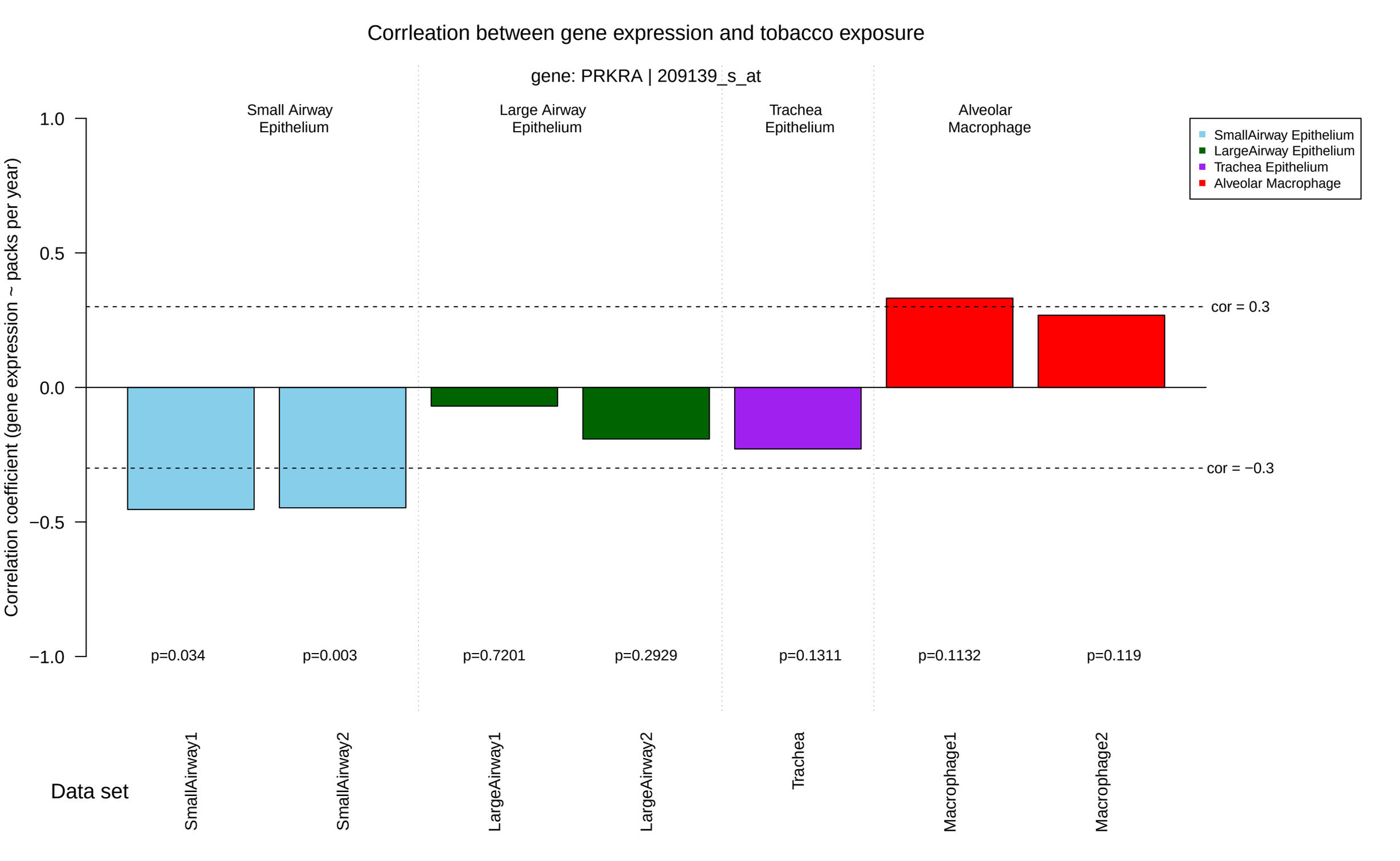

set: LargeAirway1 (p=0.0175)
PRKRA | 209139_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.069)
PRKRA | 209139_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0108)
PRKRA | 209139_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.048)
PRKRA | 209139_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1833)
PRKRA | 209139_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRKRA | 209139_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.034
p=0.003
p=0.7201
p=0.2929
p=0.1311
p=0.1132
p=0.119
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2